

Experiment No-04

Aim: Write an Assembly language program (ALP) to find Even and Odd numbers among ten numbers in memory and store the result into memory.

Software Used: Keil uVision5

Program:

```
                AREA file05, CODE, READONLY
                EXPORT __main

__main
                MOVS R0, #10
                LDR R1, =0x1FFFF000
                LDR R4, =0x1FFFF00A

LOOP
                LDRB R2, [R1]
                MOVS R5, #1
                RORS R2, R5

                BCS ODD

EVEN
                MOVS R3, #0
                B    STORE

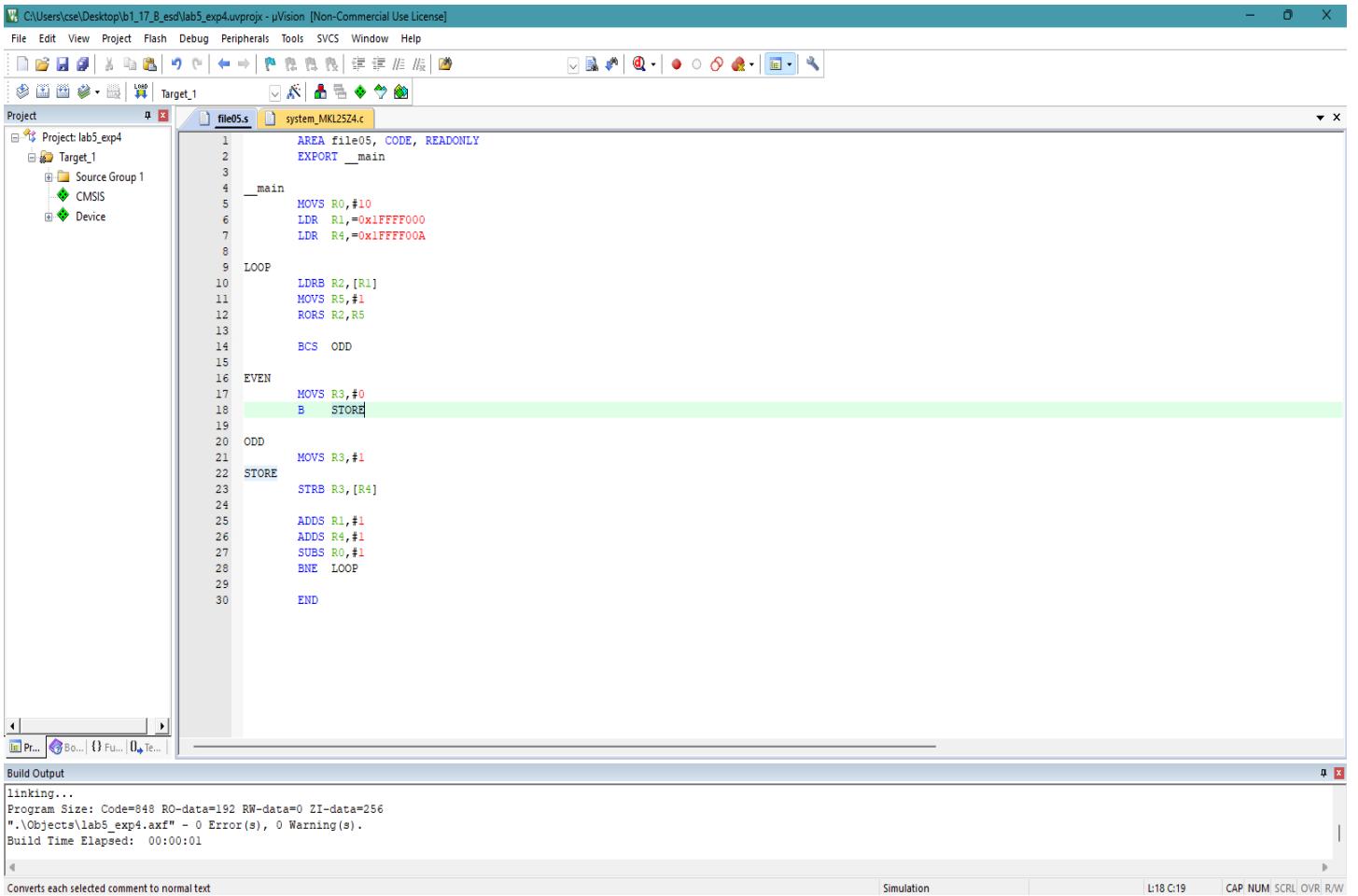
ODD
                MOVS R3, #1

STORE
                STRB R3, [R4]

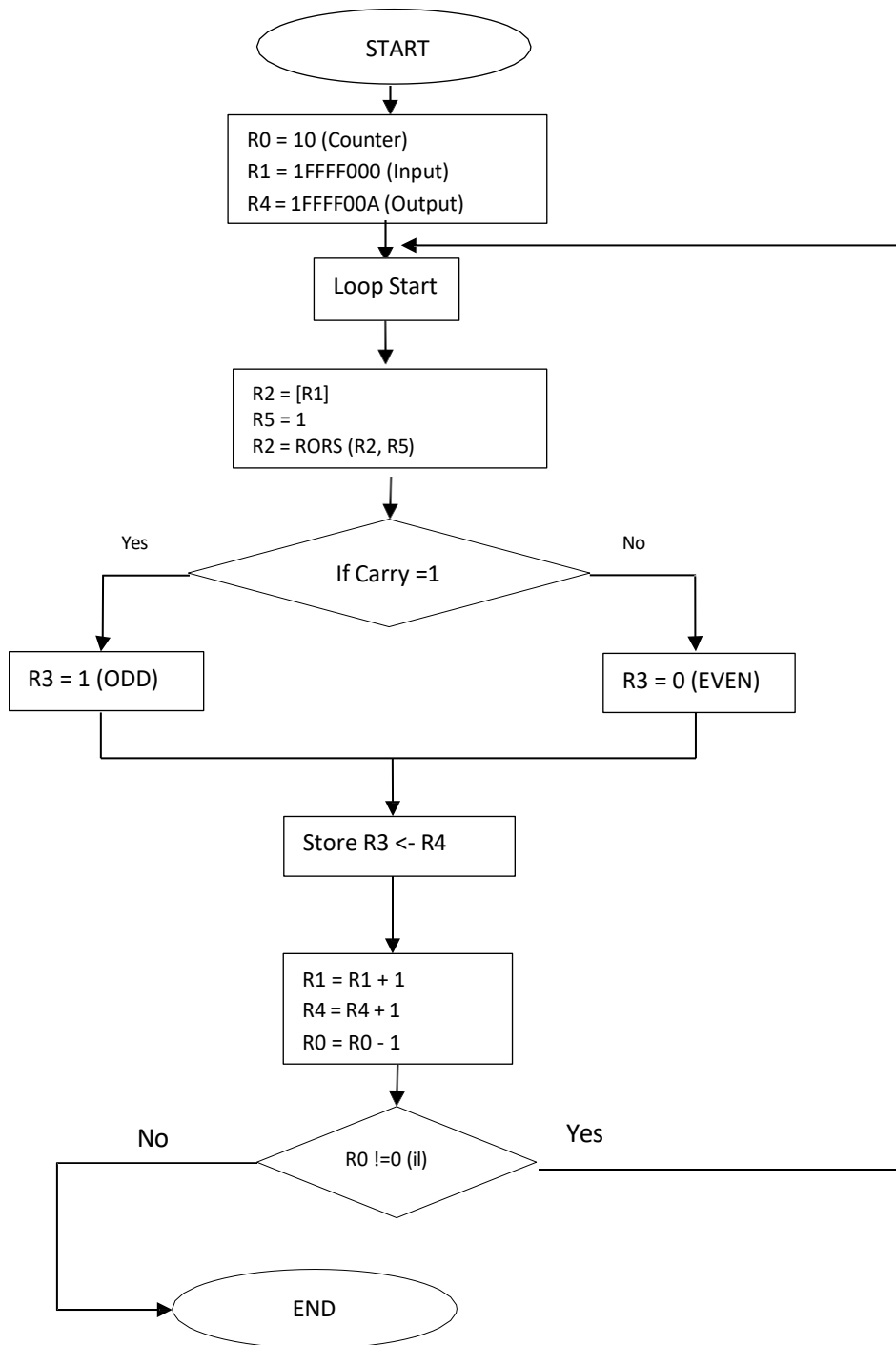
                ADDS R1, #1
                ADDS R4, #1
                SUBS R0, #1
                BNE LOOP

END
```

Program Screenshot:



Flowchart:



Results and Conclusion:

Output:

Register	Value
Core	
R0	0x00000000
R1	0x1FFFF00A
R2	0x80000011
R3	0x00000001
R4	0x1FFFF014
R5	0x00000001
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x1FFFF100
R14 (LR)	0x000000C7
R15 (PC)	0x00010000
+ xPSR	0x61000000
+ Banked	
+ System	
- Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	32784
Sec	0.00273200

Memory:

[illegible]

10 Numbers are arranged in Even or Odd and is available in Register R1 and Memory